HDOC Day -15

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Batch- 15

1. Algorithm

1. Start.
2. Input a positive integer `number`.
3. If `number <= 0`, display an invalid input message and stop.
4. Display the menu:
   - `1` — Check whether all digits are even.
   - `2` — Check whether all digits are prime.
5. Input the user's choice.
6. Use a `switch-case` statement.
7. **For Choice 1 — All digits even:**
   - Set `allEven = 1`.
   - Extract each digit using `digit = n % 10`.
   - Check `digit % 2`.
   - If any digit is odd, set `allEven = 0` and stop checking.
   - Otherwise, remove the last digit using `n = n / 10` and continue.
   - If `allEven == 1`, display "**All digits are even**"; otherwise display "**All digits not even**".
8. **For Choice 2 — All digits prime:**
   - Set `allPrime = 1`.
   - Extract each digit using `digit = n % 10`.
   - A digit is prime only if it is `2`, `3`, `5`, or `7`.
   - If any digit is not one of these, set `allPrime = 0` and stop checking.
   - Otherwise, remove the last digit and continue.
   - If `allPrime == 1`, display "**All digits are prime**"; otherwise display "**All digits are not prime**".
9. For any other menu choice, display "**Invalid choice**".
10. Stop.

2. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Number = `24680`, Choice = `1`	All digits are even	All digits are even	**Pass**
2	Number = `12345`, Choice = `1`	All digits not even	All digits not even	**Pass**
3	Number = `2357`, Choice = `2`	All digits are prime	All digits are prime	**Pass**

4	Number = 2379, Choice = 2	All digits are not prime	All digits are not prime	**Pass**

3. C Program

```
#include <stdio.h>
int main(void)
{
    long long number, n;
    int choice, digit;
    int allEven = 1, allPrime = 1;
    printf("Enter a positive integer: ");
    scanf("%lld", &number);
    if (number <= 0)
    {
        printf("Invalid input. Please enter a positive 
integer.\n");
        return 0;
    }
    printf("\nMenu:\n");
    printf("1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    n = number;
    switch (choice)
    {
        case 1:
            while (n > 0)
            {
                digit = n % 10;
                if (digit % 2 != 0)
                {
                    allEven = 0;
                    break;
                }
                n = n / 10;
            }
            if (allEven)
                printf("All digits are even\n");
            else
                printf("All digits not even\n");
            break;

        case 2:
```

```c
            while (n > 0)
            {
                digit = n % 10;

                if (!(digit == 2 || digit == 3 ||
                      digit == 5 || digit == 7))
                {
                    allPrime = 0;
                    break;
                }

                n = n / 10;
            }

            if (allPrime)
                printf("All digits are prime\n");
            else
                printf("All digits are not prime\n");

            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

```
UW PICO 5.09                    File: day15.c                    Modified

#include <stdio.h>

int main(void)
{
    long long number, n;
    int choice, digit;
    int allEven = 1, allPrime = 1;

    printf("Enter a positive integer: ");
    scanf("%lld", &number);

    if (number <= 0)
    {
        printf("Invalid input. Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMenu:\n");
    printf("1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    n = number;

    switch (choice)
    {
        case 1:
            while (n > 0)
            {
                digit = n % 10;

                if (digit % 2 != 0)
                {
                    allEven = 0;
                    break;
                }

                n = n / 10;
            }

            if (allEven)
                printf("All digits are even\n");
            else
                printf("All digits not even\n");

            break;

        case 2:
            while (n > 0)
            {
                digit = n % 10;

                if (!(digit == 2 || digit == 3 ||
                      digit == 5 || digit == 7))
                {
                    allPrime = 0;
                    break;
                }

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text    ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text  ^T To Spell
```

```
  UW PICO 5.09                                File: day15.c

                digit = n % 10;

                if (digit % 2 != 0)
                {
                    allEven = 0;
                    break;
                }

                n = n / 10;
            }

            if (allEven)
                printf("All digits are even\n");
            else
                printf("All digits not even\n");

            break;

        case 2:
            while (n > 0)
            {
                digit = n % 10;

                if (!(digit == 2 || digit == 3 ||
                      digit == 5 || digit == 7))
                {
                    allPrime = 0;
                    break;
                }

                n = n / 10;
            }

            if (allPrime)
                printf("All digits are prime\n");
            else
                printf("All digits are not prime\n");

            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

4. Compilation

```
[aaryanrajput@aaryans-macbook ~ % nano day15.c
[aaryanrajput@aaryans-macbook ~ % clang day15.c -o day15.c
 aaryanrajput@aaryans-macbook ~ %
```

Compilation result: ✅ Successful — no compilation errors.

5. Execution

Test 1 — All digits even

```
[aaryanrajput@aaryans-macbook ~ % ./day15.c
 Enter a positive integer: 24680

 Menu:
 1. Check whether all digits are even
 2. Check whether all digits are prime
 Enter your choice: 1
 All digits are even
 aaryanrajput@aaryans-macbook ~ %
```

Result: PASS ✅

Test 2 — Not all digits even

```
aaryanrajput@aaryans-macbook ~ % ./day15.c
Enter a positive integer: 12345

Menu:
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits not even
aaryanrajput@aaryans-macbook ~ %
```

Result: PASS ✅

Test 3 — All digits prime

```
aaryanrajput@aaryans-macbook ~ % ./day15.c
Enter a positive integer: 2357

Menu:
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime
aaryanrajput@aaryans-macbook ~ %
```

Result: PASS ✅

Test 4 — Not all digits prime

```
aaryanrajput@aaryans-macbook ~ % ./day15.c
Enter a positive integer: 2379

Menu:
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are not prime
aaryanrajput@aaryans-macbook ~ %
```

Result: PASS ✅